Dissertation

[TODO: create cover, add boilerplate stuff]

Acknowledgements

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Table of Contents

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Table of contents

Samenvatting

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Introduction

What is this dissertation about?

- Algorithms are everywhere.
- How can we know when to trust the output of algorithms?
  - Theoretical properties: analytically (often impossible)
  - Empirical performance: via simulation studies ('in silico'; the standard, but not standardized)
  - In practice/applied use: evaluate output, sensitivity analyses, ...? (very ad hoc, not systematic)
- In this dissertation, I investigate how the quality of the output of one specific algorithm can be evaluated.
- The algorithm central to this dissertation is MICE: multivariate imputation by chained equations. The MICE algorithm is developed to generate 'imputed values': synthetic substitutes to replace missing values in incomplete data, based on multivariate associations.
- MICE is an iterative algorithm. It produces imputations on a variable-by-variable basis. After a full pass through the data, the algorithm cycles back to the first column to replace the imputed value from the first iteration with an updated entry.

> **TODO: verwerken**
>
> Kern: waarheid. qua zeitgeist kun je wat met truth. in sim study kun je waarheid maken, maar in de praktijk moet je aannames doen. die aannames zijn belangrijk, juist omdat ze niet/ten dele te bewijzen zijn. –> waarheid vs belief. dat is waar stat als veld zich bevindt. waarheid en onzekerheid. stat = deel van de wiskunde waar we geloof nodig hebben. zonder prob desn function kune je bewijzen. je hebt het toch nooit goed, dus onderhoek hoe fout je zit. niet angst over algos.
> start met anekdote/exemplarisch verhaal.

> (**Gerko?**): please review

Missing data problems are timeless, but technological advances have made treating missingness easier than ever. A single line of code can generate several completed versions of an incomplete dataset, which allows data analysts to handle the completed datasets as if they were completely observed.

This practice of replacing missing values with synthetic substitutes is called 'multiple imputation'. Multiple imputation is a popular and flexible splits the missing data problem from the analysis of scientific interest. Imputation allows data analysts to first fill in the missingness, to then be able to analyze the completed data as if they were completely observed. So, the nuisance of missing data is conveniently resolved through imputation.

Although imputation is widely supported with software, and many implementations of the method have convenient defaults, generating *good* imputations is very hard still. In this dissertation, I set out to investigate what is needed to develop, apply, and evaluate the popular missing data method 'MICE'.

> TODO: add image for missingness/imputation?

What is MI?

- MI = replace each missing value with a synthetic substitute multiple times.
- For each missing datapoint, we draw samples from a distribution of values that *might have* been (posterior predictive distribution).
- this generates several complete*d* datasets.
- imputed datasets can be analyzed as if the data were complete.
- obtain estimates of the analysis of scientific interest on each imputation.
- estimates are then combined to obtain one pooled estimate.
- the variance of the estimate incorporates not just the sampling variance (within each imputation) but also added variance due to non-response (between imputation variance and simulation error).
- the variability between imputations reflects uncertainty due to the missingness.

> TODO: add image MI workflow

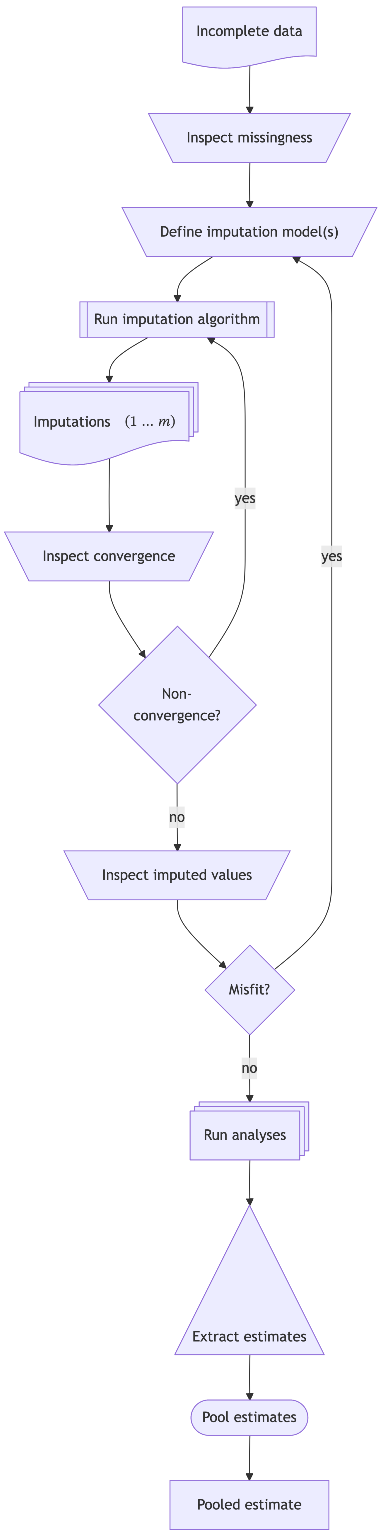
What is FCS?

- In general, imputation methods are developed to solve missing data problems.
- What types of missing data problems are we considering in this dissertation? Incomplete samples where some—but not all—entries are missing per row or per column. Entirely missing rows would be 'unit non-response'; entirely missing columns would be unobserved variables. We are interested in data with incomplete cases/variables, or 'item non-response'.
- Missing observations in incomplete cases/variables can be filled in using an imputation model
- In the FCS framework (Van Buuren et al., 2006), we need an imputation model for each incomplete variable

- fcs is substantive model agnostic, and does not assume any specific joint distribution.
- while "Rubin (Ch. 5) distinguished three tasks for creating imputations under an explicit model: the *modeling* task the *imputation* task and the *estimation* task." (van Buuren, 2007, p. 221), FSC does not require these explicit tasks. Instead, FCS only requires an *imputation model* to describe how synthetic values may be generated, without explicit specification of the joint model.
- imputation models consist of the functional form of the model (e.g., stochastic regression) and imputation model predictors (i.e., other variables in the data)
- recurring questions: how do we choose an appropriate imputation model? and how do we know whether we have chosen an appropriate imputation model?
- "Imputation models bypass the need to specify $P(Y, X, R)$, though their use creates new responsibilities for substantiating its correctness for a given statistical analysis." (van Buuren, 2007, p. 222).
- That's where computational evaluation comes in!

FCS =/= SMC

- the quality of the imputation model can only be determined/assessed based on the complete data model –> TODO: specify terminology!! –> comp eval also possible without analysis model vs stat eval relies on known analysis model (prangender vandaag want imp gebruikt voor meer analyses) fcs is substantive model agnostic
- what if that model is not (yet) available? then take the most complex possible model and check that, because if you would take the simplest model, then that would implcitly mean you are holding/constraining parameters (forcing b=0), systematically suppressing relations.
- another option would be like smc-fcs: we assume a complete data model and derive the imputations from there.


Incomplete data
Inspect missingness
Define imputation model(s)
Run imputation algorithm
Imputations (1 … m)
Inspect convergence
Non-convergence?
yes
no
Inspect imputed values
Misfit?
yes
no
Run analyses
Extract estimates
Pool estimates
Pooled estimate

TODO: verwerken

misfit en non-conv next to each other
Computational evaluation is in every trapezoid: processes that require human intervention.
Statistical evaluation is mainly focused on the relation between the pooled estimates and
Paragraph after this: convince reviewers that this is the right choice. Giswerk/gebrek aan controle leidt to suboptimale estimates
missing: draw inference
tussen inspect miss en def imp mods: involve all stakeholders crips dm (data mining)
–> pre vs post imp facilitator functies om te checken of er nieuwe verbanden in zitten (kruistabel)

What is computational evaluation?

- Computational evaluation is not equal to statistical evaluation. Statistical evaluation is important, e.g. to quantify bias and variance of estimators/procedures, but not the focus of this dissertation.
- Computational evaluation is something to check *conditional on* acceptable statistical properties, and does not replace statistical evaluation.
- Computational evaluation is originally from information retrieval research (see e.g. "Evaluation Measures (Information Retrieval)," 2025)
- The practice of computational evaluation combines *effectiveness* (i.e., did the process yield the correct result, e.g., accuracy, precision, recall, etc.), *system quality* (e.g., speed, coverage of all possible results, etc.) and, importantly, *user utility* (e.g., happiness, productivity, etc.) (Manning et al., 2008)
- The term 'computational evaluation' is not specifically defined for the field of statistics, but it could be useful.
- One definition of computational evaluation is "to determine the quality of solutions attainable" (Dudek, 2004)
- Translated to statistics, a model may be statistically valid, but not fit for purpose. For example, computations that take too long to converge for use in a production pipeline/real-time use in patient care.
- Another definition of computational evaluation is "the process of ensuring an algorithmic solution is a good one: that it is fit for purpose" (Curzon et al., 2014).
- We can use this when a statistical process has different purposes and thus different usability/requirements. For example, calculating uncertainty at the level of a single case, a sample, or inferring to a population.
- Since the term is not defined for statistics, there is no definition to use for the computational evaluation of imputation methodology.
- In this dissertation, I want to plant a flag: how can we define 'computational evaluation' in the context of imputation methodology?
- Working definition: *systematic assessment of the algorithmic behavior and data-level*

outputs of imputation procedures, focused on: algorithmic stability (convergence, failure modes), sensitivity to modeling choices, plausibility and coherence of imputed values, and usability for applied researchers.

- These aspects are often already taken into account by imputers, but I try to systematize and operationalize these aspects that are often informal or *ad hoc*.

Why should we care about computational evaluation of imputation methodology?

- Computational evaluation goes beyond statistical evaluation.
- Even if we know the process yields correct inferences, we can evaluate the usability/appropriateness/plausibility of the solution. A question to consider is: do the imputed values lie within the range of the observed values?
- Imputation theory does not require the imputed values to be plausible.
- Rubin (1976, 1987) developed the methodology to get correct population-level inferences from incomplete samples, not at the individual case-level within the sample.
- Bluntly stated: imputation pragmatists/purists don't care for the imputed values themselves, as long as they result in statistically valid inferences.
- But what are the limitations of purely inferential metrics (bias, coverage)? e.g. imputing negative values for age, using PMM for a variable with a sparse region (yielding 'unrealistic' imputed values), or evaluating imputed values of non-converged chain (?), or not knowing the analysis model up front so congeniality is at risk (???) -> PPC would show problem with sparse region
- And vice versa: when might the plausibility of imputed values mislead a data analyst into believing the imputations are statistically valid? when 'looking reasonable' is confused with 'being consistent with the data-generating process', such as single imputation (stoch reg), or imputing under MCAR assumption when there is a MAR mechanism, or destroying the joint distribution of the data while preserving the marginals. TODO: figure out the difference with Gerko's interpretation of joint occurrence (cross tabs) within conditional distribution; should we make an algorithm to show user which combinations of bivariate obs only occur in imp?
- Statistical evaluation of bias and coverage is not possible in practice, only in simulation studies.
- Stat evalualtion may not possible IRL? Bias and confidence interval coverage are impossible to determine without a comparative truth (although PPC and bootstrapping allows us to compare with respect to an estimand).
- Computational evaluation means not just taking the answer as-is, but tracing back the solution of an algorithm to its origin/back-tracing the procedure: you should *always* be able to retrieve the chain of the imputation procedure. E.g., implausible or impossible imputed values can yield valid statistical inference, but are we satisfied with the way that the imputation were obtained?
- What do we care about? bias/coverage/variance, empirical relevance/plausibility, software-engineering nitpicks like speed/convergence/fault rate.

- In this dissertation I will use computational evaluation as a suite of tools to investigate the domains of algorithmic stability, data plausibility, and user burden, next to inferential validity.

> **Examples of stat but not comp**
>
> - polynomials (ignoring the U in imps could still get stat val)
> - example: psych study with baseline, some of which are missing; baseline, intervention, follow-up –> mice would inflate baseline and follow-up if both are incomplete; stat eval will not show this (unbiased, cov ok, etc.); over-imp or postpred check would show diff distr for obs and imp; circularity should be removed: follow-up should not predict baseline
> - fcs iterations not stable, but estimand ok (in sim study true value) –> conv at cell level??
> - PMM with part of sample space with too little donors

What is the scope of this dissertation?

RQ: what tools are required to assess the quality of imputation methods? in the development, evaluation, and application.

- is the status quo sufficient? no, stat eval is required, but not sufficient.
- how Rubin developed the methodology works in theory, but is not enough for practitioners
- can we use existing tools or do we need more? and how? no, not all are present in the imputor's toolbox. I contribute to the new status quo ("verleggen" = generalizable scientific contribution of this dissertation)
- unifying how imputations are generated and evaluated (??)
- lifecycle of imputation methods: method is theorized, method is implemented in software, method is evaluated in simulation study, method is put to scrutiny by scientific community, method is applied in practice, method is evaluated in real-world data.
- two types of evaluation: statistical properties of the method in simulation (e.g., bias and confidence-validity; statistical evaluation) & practical application (computational evaluation).

In this dissertation, we will focus on missing data problems with item non-response and MCAR/MAR missingness mechanisms. Unit non-response and MNAR missingness mechanisms are outside of the scope of this dissertation. The missing data method under consideration is primarily multiple imputation by chained equations (MICE), but some results may generalize to other iterative (FCS) imputation algorithms. We do not cover joint modeling (JOMO) imputation or non-imputation missing data methods. Although MICE is not SMC-FCS, we will restrict the analysis of scientific interest mostly to GML family of models. The applications are aimed at statistical inference (not prediction or causal inference).

Introducing chapter 1: `ggmice`

This chapter introduces the software ‘ggmice’: an R package for building and evaluating imputation models. The ‘ggmice’ package offers tools for the visualization of incomplete data, imputation models, and imputed data. ‘ggmice’ is compatible with the popular R packages ‘mice’ (for imputation) and ‘ggplot2’ (for flexible data visualizations based on the ‘grammar of graphics’ framework). This fills a hiatus in the software landscape, with other packages generally supporting either ‘mice’ or ‘ggplot2’, but not both. Table REF provides an overview of R packages that offer visualization tools for incomplete and imputed data. In short: with `mice` you cannot plot incomplete data distributions, all other packages lack support for `mice`-imputed data, and visualization of ‘mice’ imputation models did not exist at all prior to ‘ggmice’.

	ggmice	mice	naniar	VIM
Incomplete data (e.g., `data.frame` object)				
– joint & marginal distributions	+	–	+	+
– missing data patterns	+	+	±	+
– imputation models	+	–	–	–
Imputed data (e.g., `mids` object)				
– joint & marginal distributions	+	+	–	–
– imputation models	+	–	–	–
– trace plot of the imputation algorithm	+	+	–	–
User utility (e.g., `ggplot` object)				
– flexibility/extensibility of visualization types	+	–	±	–
– adaptability/publication readiness of visualizations	+	–	+	–

> TODO: write table footnote to explain all aspects of the table. i.e., these are only diffs. the table does not include things that all packages can –> maybe as appendix?
> also add number of downloads: `ggmice` has 800 monthly compared to `VIM` 13k and `naniar` 22k

When would you need viz tools?

- Model building: how do we know which functional form to choose?
  - rely on defaults, tested in simulation studies
  - assess the distribution of the incomplete variable (e.g., visual inspection)
  - …? (maybe ad transformations?)
- Model building: how do we know which imputation model predictors to select?

  - association of incomplete variable with other variables in the data ("include an auxiliary variable if its correlation with the main variable is 0.5 (in absolute value) [25]" Nguyen et al., 2021)
  - association of missingness indicator with other variables in the data
  - external input (e.g., branching patterns, or expert knowledge to correct for MNAR)
  - …?

- Model evaluation: how do we assess imputation model misfit?

  - non-convergence in the algorithm
  - mismatch between observed and imputed data distributions (e.g., visual inspection)
  - …?

What does ggmice aim to solve?

The 'ggmice' package aims to solve the following problems:

- no direct graphical comparisons between incomplete and imputed data; side-by-side presentation of the data before and after imputation; Visualizations with 'ggmice' can be made to inspect marginal and joint distributions of incomplete variables side-to-side to compare incomplete and imputed data
- no previously existing methods for visualizing `mice` imputation models; complex imputation models are easier to review through visualization (as opposed to console prints/excel exports of predictor matrix)
- existing plotting tools for `mids` objects not easily editable/publication-ready; Visualizations with 'ggmice' can be extended and adapted with tidyverse functions to create publication-ready graphs of imputed data (reflecting the uncertainty due to missingness).
- unified toolbox for the development and evaluation of imputation models

Design principles, assumptions and limitations

- Open Source Software development/FAIR –> R, GitHub, CRAN, Zenodo, …

- Grammar of Graphics: layered plots (Wilkinson, 2012) –> - `ggmice` produces `ggplot` objects: layered, easily adjustable

- evaluation of the distribution of observed and imputed data –> imputation model should fit the observed part, imputation model should be congenial –> providing tools to see observed and imputation side-by-side

- make explicit what these plots/analyses can and cannot tell us: imputation models should fit the observed part of the data, but no way to determinately state miss mech.

- visualization =/= objective truth –> good imputation model fit/plausible values is not a guarantee for statistical validity

- visualization is not a replacement of evaluation but support, also check metrics such as FMI
- interpretation remains subjective (e.g. convergence, MAR assumption, etc.)

Future research & development

- visualizing `mira` objects
- add posterior predictive check plots & over-imputation
- quantifying uncertainty at pattern/cell level
- tools for sensitivity analyses
- What data-level diagnostics are informative for assessing imputation model adequacy in practice (i.e., when inferential validity cannot be evaluated)?
- Do we get better imputations in practice?

Indicators for adoption

- software published on Zenodo (Oberman, 2022), with open code and bug reporting via GitHub
- peer-review not formal via journal, but via Issues and Pull Requests on GitHub (10x external Issue, 13x external PR)
- impact: citations (5x according to Google Scholar), CRAN downloads (852x/month; 21k total) and conda downloads (100+/month; https://anaconda.org/channels/conda-forge/packages/r-ggmice/overview)
- CRAN downloads
- GitHub stars and issues
- StackOverflow mentions/questions
- feedback at conferences
- citations of the package (Zenodo reference)

Introducing chapter 2: convergence

- **RQ: FSC is iterative, so convergence *should* be a requirement for valid inference, or is it?**/How does non-convergence in iterative imputation affect inferential validity, and how can non-convergence be diagnosed?
- simulation study, published as pre-print (11x cited according to Google Scholar), presented at ICML

- even without convergence, we can get valid inference at sample/population level, but not at individual cell level
- imputation stays a custom job: convergence of imputation is conditional on imputation model
- today, imputation are not generated for 1 analysis prob, but all analyses that may be fitted on the data –> all imputation models should be compatible and congenieal –> more 'omvattende' imputation models required
- liefst wil je analysemodel parameters tracken, daarom is computational evaluation zo belangrijk

Introducing chapter 3: evaluation

- **RQ: what to look out for when designing simulation studies for the evaluation of imputation methodology?**/How can computational evaluation criteria be systematically incorporated into simulation studies for imputation methodology?
- published as Oberman & Vink (2024), cited 27x according to Google Scholar (of which 11x CrossRef)
- based on literature review Liu et al. (2023)

Chapter 1: ggmice

[TODO: link/insert]

Chapter 2: convergence

[TODO: link/insert]

Chapter 3: evaluation

[TODO: link/insert]

Discussion

Conclusion

- Computational evaluation of imputation methodology is never finished.
  - In practice, you cannot validate whether the imputation model was appropriate (???), because what you would need to do so is exactly the thing you don't have: the missing part of the incomplete data, while you only have the observed part.
  - The best thing we can do is to rely on a combination of assumptions/theoretical justifications/substantive expertise and indirect evidence of imputation model fit: check the observed versus imputed data distributions, and inspect the imputations for signs of non-convergence. This dissertation offered insight and tools to do so.
  - The only setting where we do have the comparative truth (the true but missing values) is simulation studies. Imputers have to rely on simulation study results in order to choose the best imputation method for their incomplete variables. That's why simulation studies for imputation methodology should be comparable with one another, so imputers can make a 'grounded' assessment which imputation methods may be applicable, and choose the most appropriate one.

–> example: NRI search for lower bound, so impute as 'positive behavior', but per definition wrong statistical inf, because we intentionally bias in one direction

Implications

- The main findings in this thesis are…
  - Non-convergence is hard to diagnose, and typical thresholds to evaluate non-convergence may not be applicable to FCS algorithms: MICE reaches stable output before non-convergence metrics would indicate so. There is a mismatch between theoretical convergence criteria and practical algorithmic behavior.
  - Visual inspection aids imputers in developing and evaluating imputation models, because formal tests are not available. The R package `ggmice` offers tools for the computational evaluation of the imputations models that were previously unavailable.
  - Defining and testing new imputation methods through simulation studies requires careful consideration of the simulation design and evaluation to make imputation methods comparable across studies.
- Recommendations for fellow missing data methodologists:
  - design simulation studies structured and reproducible (e.g., report design choices, publish code)
  - talk to your intended audience (i.e., applied researchers), or at least check what they are struggling with (e.g., StackOverflow)

  - …?

- Recommendations for fellow imputers:
  - think before you code
  - look at the data
  - …?

Limitations

- Computational evaluation is not a substitute for thinking. It may give a false sense of certainty, e.g. against MNAR mechanism (untestable assumption).
  - Use expert insight: use computational evaluation as complement to (not a replacement for) substantive knowledge and sensitivity analyses
  - Take the analysis model (if available) as starting point for imputation workflow, for congeneality
  - …?
- This entire dissertation relies on the R language and centers the `mice` package (TODO rephrase as not a flaw: I focus on a dominant ecosystem, and concepts generalize beyond).
  - While this set-up is very popular, the data science landscape has expanded and is ever evolving
  - Many machine learning and deep learning methods are not (yet) implemented as imputation models in `mice`. These might be able to better approximate the posterior predictive distribution of the missing values than `mice` methods, but research should show that still.
- ~~I haven't tested whether the tools I developed actually improve the imputations of `mice` users~~ This dissertation evaluates tools and diagnostics at the methodological level; assessing their causal impact on user behavior and imputation quality remains an important direction for future empirical research.
  - GitHub issues and StackOverflow are indication
  - Onmogelijk om evidence-based conclusies te trekken, maar misschien wel evidence-informed?
- The visualizations implemented in `ggmice` are generally better suited for numeric data, as opposed to categorical variables (or even open text field data, which is not supported by `mice` currently). Future research on: associations of cat vars with miss indicators, and convergence parameter other than SD of imputed values.

> - TODO: rephrase 'haven't's into future research (positive phrasing: reframe as scope conditions, not shortcomings)

- maybe future: computational evaluation of variable importance within imputation models

Outlook

- This chapter ends with a future perspective…
- Missing data remains a problem indefinitely. I expect imputation methodology to be developed and applied for the foreseeable future. But the way we deal with data might change.
- With the rise of machine learning and deep learning methods (or "AI"), evaluation will become ever more important. Novel methods are being developed under a prediction framework, not statistical inference framework: ignoring the uncertainty in the estimates, and thus not incorporating between-imputation variance. This leads to too narrow CIs, and too low p-values. Which, in turn, causes spurious results.
- AI models should propagate uncertainty!
- trust in science requires honest reflection of uncertainty in our analyses
- be open about uncertainty, and limitations of scientific studies
- publish null results (e.g., registered reports or pre-registrations)
- share data and code with other scientists, and the public
- publish open access, educational materials too!
- change the way we evaluate science: this dissertation is an example of the new recognition and rewards vision. it includes software as an actual chapter, not something extra.
- I hope to inspire others to reflect on their own view of what scientific output is. And most of all, I hope to set an example for other PhD candidates to do the same
- imp is not always the way to go: missingness can tell you about mismatch between data collection method and lived experience

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CV

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